

air_conditioner

April 10, 2025

```
[1]: !pip install numpy matplotlib scikit-fuzzy scipy networkx
```

```
Requirement already satisfied: numpy in ./venv/lib/python3.10/site-packages
(2.2.4)
Requirement already satisfied: matplotlib in ./venv/lib/python3.10/site-packages
(3.10.1)
Requirement already satisfied: scikit-fuzzy in ./venv/lib/python3.10/site-
packages (0.5.0)
Requirement already satisfied: scipy in ./venv/lib/python3.10/site-packages
(1.15.2)
Requirement already satisfied: networkx in ./venv/lib/python3.10/site-packages
(3.4.2)
Requirement already satisfied: kiwisolver>=1.3.1 in ./venv/lib/python3.10/site-
packages (from matplotlib) (1.4.8)
Requirement already satisfied: cycler>=0.10 in ./venv/lib/python3.10/site-
packages (from matplotlib) (0.12.1)
Requirement already satisfied: pyparsing>=2.3.1 in ./venv/lib/python3.10/site-
packages (from matplotlib) (3.2.3)
Requirement already satisfied: contourpy>=1.0.1 in ./venv/lib/python3.10/site-
packages (from matplotlib) (1.3.1)
Requirement already satisfied: packaging>=20.0 in ./venv/lib/python3.10/site-
packages (from matplotlib) (24.2)
Requirement already satisfied: fonttools>=4.22.0 in ./venv/lib/python3.10/site-
packages (from matplotlib) (4.57.0)
Requirement already satisfied: pillow>=8 in ./venv/lib/python3.10/site-packages
(from matplotlib) (11.1.0)
Requirement already satisfied: python-dateutil>=2.7 in
./venv/lib/python3.10/site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in ./venv/lib/python3.10/site-packages
(from python-dateutil>=2.7->matplotlib) (1.17.0)
```

0.1 Import libraries

```
[2]: import numpy as np
import matplotlib.pyplot as plt
import skfuzzy as fuzz
from skfuzzy import control as ctrl
```

0.2 Universe variables

```
[3]: temperature = ctrl.Antecedent(np.arange(0, 51, 1), 'temperature')
      humidity = ctrl.Antecedent(np.arange(0, 101, 1), 'humidity')
      fan_speed = ctrl.Consequent(np.arange(0, 101, 1), 'fan_speed')
```

```
[4]: temperature
```

```
[4]: Antecedent: temperature
```

```
[5]: temperature.universe
```

```
[5]: array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
          17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
          34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50])
```

```
[6]: humidity
```

```
[6]: Antecedent: humidity
```

```
[7]: humidity.universe
```

```
[7]: array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12,
          13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25,
          26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38,
          39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51,
          52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64,
          65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77,
          78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90,
          91, 92, 93, 94, 95, 96, 97, 98, 99, 100])
```

```
[8]: fan_speed
```

```
[8]: Consequent: fan_speed
```

```
[9]: fan_speed.universe
```

```
[9]: array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12,
          13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25,
          26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38,
          39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51,
          52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64,
          65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77,
          78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90,
          91, 92, 93, 94, 95, 96, 97, 98, 99, 100])
```

0.3 Create membership functions

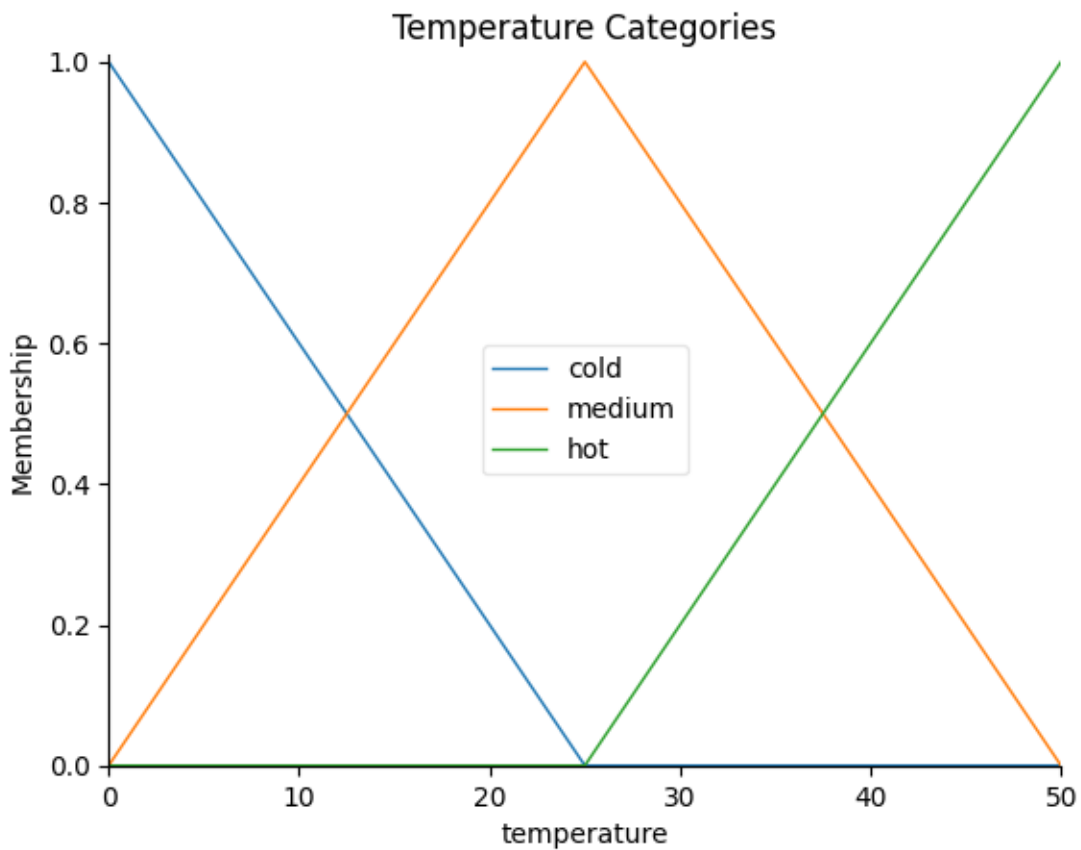
```
[10]: temperature.automf(3, names=['cold', 'medium', 'hot'])
humidity.automf(3, names=['low', 'medium', 'high'])

fan_speed['low'] = fuzz.trimf(fan_speed.universe, [0, 0, 50])
fan_speed['medium'] = fuzz.trimf(fan_speed.universe, [0, 50, 100])
fan_speed['high'] = fuzz.trimf(fan_speed.universe, [50, 100, 100])
```

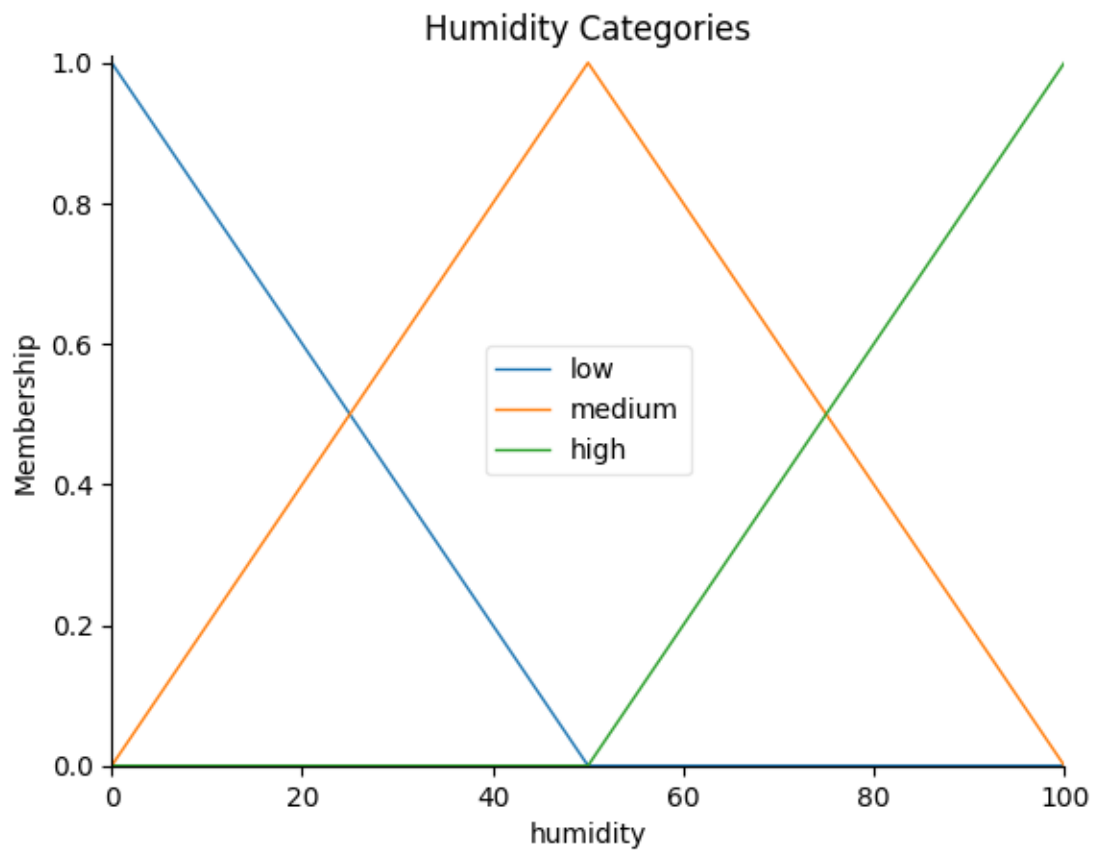
0.4 View The Functions

```
[11]: temperature.view()
plt.title('Temperature Categories')
plt.show()
```

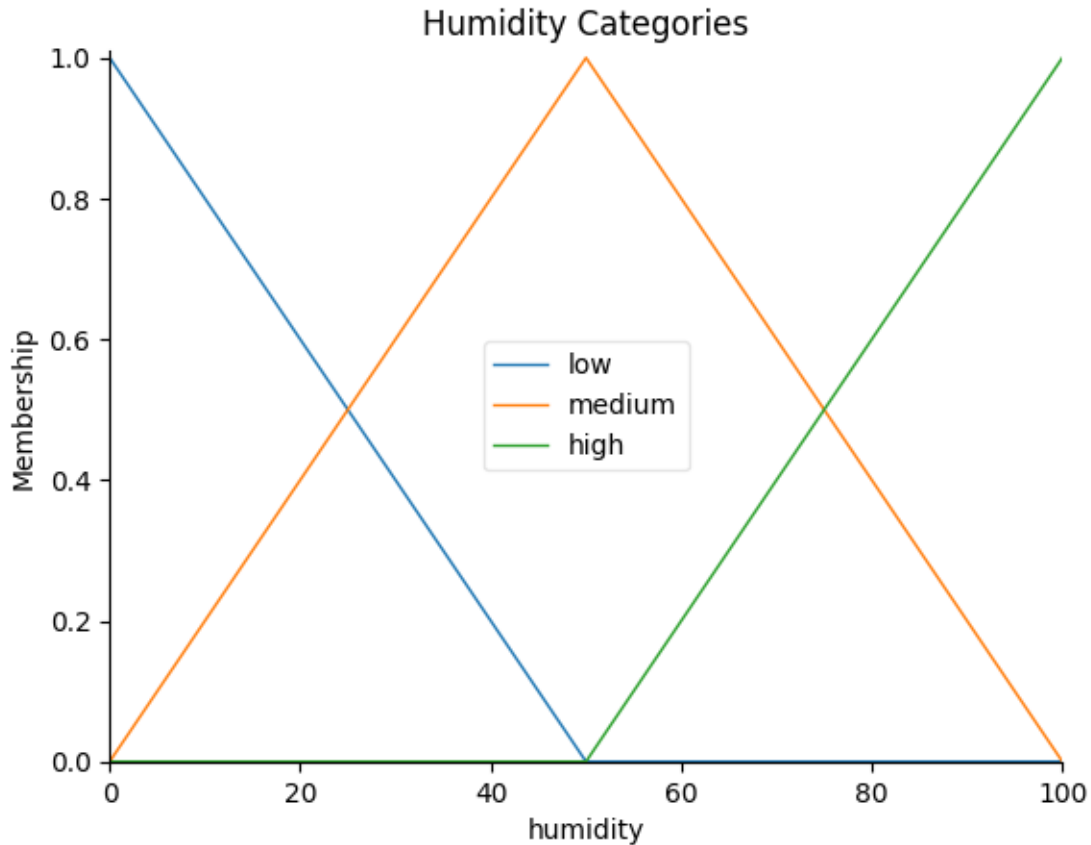
/home/stark/projects/fuzzy/venv/lib/python3.10/site-packages/skfuzzy/control/fuzzyvariable.py:125: UserWarning: FigureCanvasAgg is non-interactive, and thus cannot be shown
fig.show()



```
[12]: humidity.view()  
plt.title('Humidity Categories')  
plt.show()
```



```
[13]: humidity.view()  
plt.title('Humidity Categories')  
plt.show()
```



0.5 Define rules

- If temperature is hot OR humidity is high, then fan speed is high.
- If temperature is medium AND humidity is medium, then fan speed is medium.
- If temperature is cold, then fan speed is low.

```
[14]: rule1 = ctrl.Rule(temperature['hot'] | humidity['high'], fan_speed['high'])
rule2 = ctrl.Rule(temperature['medium'] & humidity['medium'],
    ↪ fan_speed['medium'])
rule3 = ctrl.Rule(temperature['cold'], fan_speed['low'])
```

0.6 Create control system

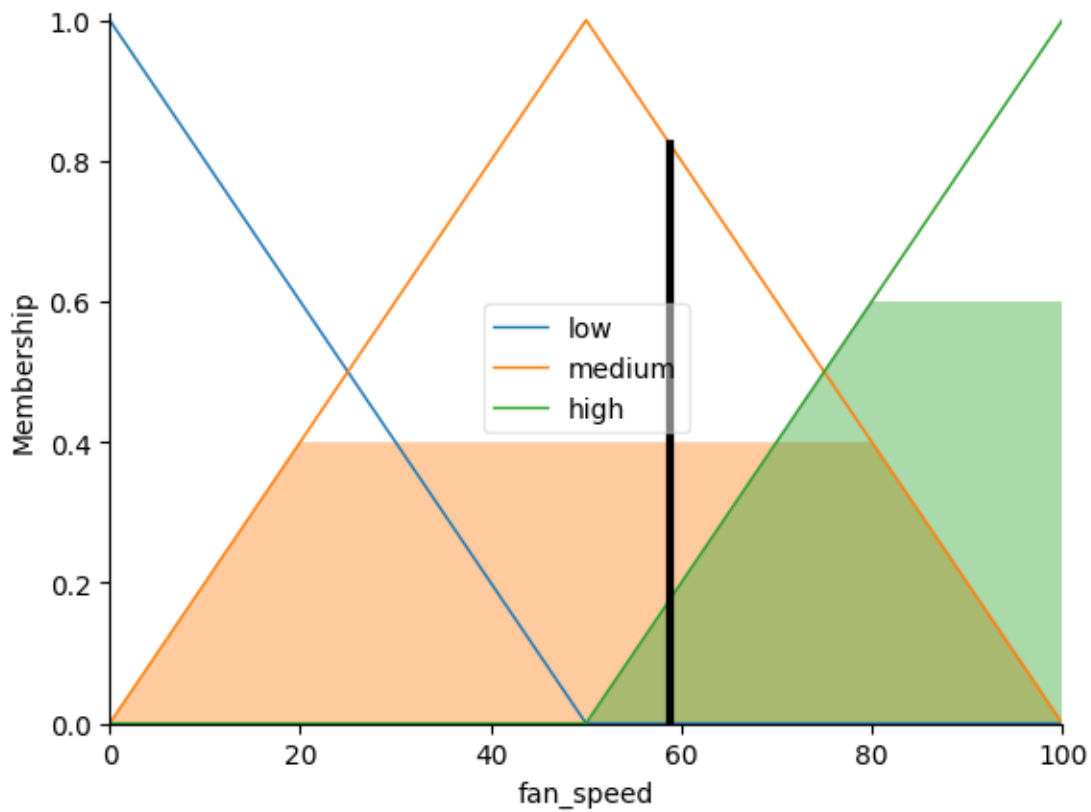
```
[15]: fan_ctrl = ctrl.ControlSystem([rule1, rule2, rule3])
fan_system = ctrl.ControlSystemSimulation(fan_ctrl)
```

0.7 Simulate the Fuzzy System

```
[16]: fan_system.input['temperature'] = 35
fan_system.input['humidity'] = 80
fan_system.compute()

print(f"Fan Speed: {fan_system.output['fan_speed']:.2f}%")
fan_speed.view(sim=fan_system)
plt.show()
```

Fan Speed: 58.78%



```
[17]: temp_hot = fuzz.interp_membership(temperature.universe, temperature['hot'].mf, 35)
humidity_high = fuzz.interp_membership(humidity.universe, humidity['high'].mf, 80)

print(f"Temperature 'Hot' Membership: {temp_hot:.2f}")
print(f"Humidity 'High' Membership: {humidity_high:.2f}")
```

Temperature 'Hot' Membership: 0.40
Humidity 'High' Membership: 0.60

[]: